

Jiyeon Shin

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EDUCATION

Georgia Institute of Technology

M.S. in Computer Science (Machine Learning)

Atlanta, GA

May 2028

Yonsei University

B.S. in Urban Planning and Engineering & Applied Statistics (GPA: 3.9/4.0)

South Korea

Feb. 2026

EXPERIENCE

Smart City Infrastructure Lab, Georgia Institute of Technology

Graduate Research Assistant

Aug. 2026 - Present

Atlanta, GA

- Built a GNN-based reinforcement learning framework to optimize traffic signals across multiple intersections
- Developed an autoencoder approach to detect anomalous vehicle trajectories associated with intersection crash risk

Kearney

Data Engineering Intern

Mar. 2026 - Jun. 2026

South Korea

- Built and maintained ETL pipelines to integrate 40M+ traditional and social media records for PR performance analytics
- Engineered LLM-based classification systems to automate analysis of 355K+ social media posts, achieving ~89% accuracy
- Automated identification of HQ articles across 40+ subsidiaries using semantic similarity, cutting review time by half

Kakao Mobility

Data Science Intern

May 2025 - Feb. 2026

South Korea

- Identified ETA error patterns through exploratory analysis of 100M+ navigation trips, informing model improvements
- Designed a metric to measure adherence to recommended routes and automated 10-minute monitoring with Airflow
- Developed a data-driven method to select historical dates for speed prediction, reducing MAE by up to 9%
- Engineered traffic-flow features and clustered roads with PCA and K-means, achieving 0.85 ARI across time periods

Transportation Planning Lab, Yonsei University

Undergraduate Research Assistant

Mar. 2024 - Jun. 2024

South Korea

- Analyzed 15K+ trips using statistical tests and stepwise regression to refine access-time measures for demand modeling
- Modeled ride-hailing wait tolerance using survival analysis to assess user sensitivity to service attributes

PROJECTS

MiniGrad: Neural Network Engine

Sep. 2026 - Present

- Implemented backpropagation from scratch with reverse-mode autodiff and dynamic computation graphs
- Built neural network layers, loss functions, and optimizers on top of the engine and validated training against PyTorch

Spatial Data Simulator

Aug. 2026 - Present

- Built a simulator to generate synthetic spatial data with controlled spatial patterns and known ground-truth relationships for evaluating computational methods

SKILLS

Languages: Python, SQL, R, Java

Data & Platforms: BigQuery, PostgreSQL, Airflow, GCP, AWS, Docker, Tableau, Git

Statistics & ML: Statistical Inference, Bayesian Modeling, Predictive Modeling, Time-Series Analysis, Transformers

PUBLICATIONS

- Ziyu Liu, ..., **Jiyeon Shin**, et al. "AD-GPT: Large Language Models in Alzheimer's Disease." *BMC Medical Informatics and Decision Making* (2026). <https://doi.org/10.1186/s12911-026-03579-x>